```python
def alphabeta(depth, node, maximizing, values, alpha, beta):
    if depth == 2:
        print("Evaluating leaf:", values[node])
        return values[node]

    if maximizing:
        best = float('-inf')
        for child in [node * 2, node * 2 + 1]:
            best = max(best, alphabeta(depth + 1, child, False,
                                       values, alpha, beta))
            alpha = max(alpha, best)
            if beta <= alpha:
                print("Branch pruned at node:", child)
                break
        return best
    else:
        best = float('inf')
        for child in [node * 2, node * 2 + 1]:
            best = min(best, alphabeta(depth + 1, child, True,
                                       values, alpha, beta))
            beta = min(beta, best)
            if beta <= alpha:
                print("Branch pruned at node:", child)
                break
        return best

values = [3, 5, 2, 9]
print("Leaf Scores:", values)
result = alphabeta(0, 0, True, values,
                   float('-inf'), float('inf'))
print("Optimal Score:", result)
```

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/AIES.py ==========
========
Leaf Scores: [3, 5, 2, 9]
Evaluating leaf: 3
Evaluating leaf: 5
Evaluating leaf: 2
Branch pruned at node: 2
Optimal Score: 3
>>>
```